David Kozák

Software Engineer

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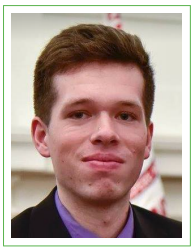

Who am I ?

I am a passionate Java/Kotlin developer who loves programming, learning new things and personal development in general. I have general knowledge of Java EE, Spring and JPA/Hibernate. I used these at work on daily basis. After some time being a pure backend developer, I decided to explore the frontend as well. Now I have experience with vanilla JavaScript(ES6 and beyond) and its frameworks such as React and Angular. As a fan of statically typed languages, I quickly moved from JavaScript to TypeScript. I am also a fan of Haskell and functional programming in general. Thanks to my internship at JetBrains, I had a chance to learn about DSL engineering and language design and implementation in general.

What am I looking for ?

If I were to describe my perfect position, it would be somewhere on the edge between theory and practice. I'd like to solve complex problems using theoretical knowledge from academia and implement the solution as an industrial tool. In particular, I am interested in compilers and code analysis tools. On top of that, I'd like my job to involve developer advocacy as well, since I enjoy public speaking, teaching, travelling and in general I like being around people.

Work Experience

Sep 2020 – now **Research Assistant**, *Oracle Labs*, Oracle, Brno.
Working on the GraalVM - a high-performance polyglot virtual machine.

Mar 2019 – Aug 2020 **Software Engineer**, *Knowledge Technology Research Group* , FIT BUT, Brno.
I work on a search engine for querying semantically enhanced documents available within Knowledge Technology Research Group at FIT BUT. It is a distributed system consisting of multiple types of components. The task also includes designing a special query language, it's compiler and specialized searching algorithms. Backend components are written in Kotlin using Spring. Web frontend is written in TypeScript using React. My responsibility is to design, develop and deploy the whole system.

Sep 2018 – Feb 2019 **Software Developer**, *MPS Team*, JetBrains s.r.o., Prague.
I dived deeper into language engineering using MPS and learned how to use it to develop DSLs and IDE support for them.

Jul 2018 – Aug 2018 **Internship**, *MPS Team*, JetBrains s.r.o., Prague.
I studied DSL engineering and language engineering in general. In particular, I worked on a tool for converting Antrl4 grammar into MPS project. I designed and implemented an extension to support automatic detection of formatting for a given language.

Feb 2018 – Aug 2018 **Java Full Stack Developer**, *Aura*, Aura s.r.o., Brno.
I worked on an information system using JavaEE, Spring and Hibernate. I used JSF on the presentation layer and Spring and Hibernate in the backend.

Jun 2015 – Dec 2017 **Java Developer in Research**, *Automated Analysis and Verification Research Group - VeriFIT*, FIT BUT, Brno.
I maintained SearchBestie infrastructure for search-based testing of concurrent Java programs using noise injection. My main task was to integrate SearchBestie with RoadRunner, an open source tool for dynamic analysis of concurrent Java programs. I also designed new heuristics for noise injection and evaluated them.

Jan 2015 – Dec 2015 **Android Developer**, *BeeeOn*, FIT BUT, Brno.
I worked on an Android application for controlling an IoT system. First, I was focused on the development, then I moved towards testing. I prepared an automated testing support for the app and integrated it with Jenkins.

Education

2017 – now **Brno University of Technology**, *Master of Computer Science*, *Information Systems*.

Feb 2019 – Jul 2019 **Escola Superior de Tecnologia e Gestao**, *Erasmus student*, *Software engineering*.

Sep 2018 – Jan 2019 **University of Southern Denmark**, *Erasmus student*, *Software engineering*.

Feb 2017 – Jul 2017 **Technological Educational Institute of Crete**, *Erasmus Student*, Heraklion, Crete, Greece.

2014 – 2017 **Brno University of Technology**, *Bachelor of Computer Science*, *Information Technology*.

2010 – 2014 **Grammar School, Svitavy**.

Bachelors Thesis

Title *Fine-Grain Noise-Injection Heuristics for SearchBestie Infrastructure*

Supervisors Ing. Bohuslav Křena, Ph.D.

Description Testing and runtime verification of concurrent Java programs.

Volunteering

Sep 2019 – Dec 2019 **Vicepresident**, *ESN VUT*, ESN CZ.
On top of my normal responsibilities as an ESN member, I also organized and moderated our regular meetings, took care of our office and helped our president with running the student club.

Jul 2017 – Sep 2019 **ESN member**, *ESN VUT*, ESN CZ.
I helped foreign students during their Erasmus in Brno. For example, I taught Czech lessons and organized events for hundreds of Erasmus students. I used English on every day basis and learned to communicate and cooperate with different types of people from different cultures and backgrounds.

Skills

General Problem solving, algorithms and data structures, software design and architecture, design principles and patterns.

Languages Kotlin, JavaScript/TypeScript, Java, Python.

Frameworks and libraries React & Redux, JavaFX, Java EE, JPA/Hibernate, Spring.

Interest Compilers, interpreters, code analysis, DSL Engineering, IDE development.

Soft skills Public speaking, teaching, teamwork.

Achievements

Apr 2018 **UnIT Hackaton - Webapps**, *third place*.

Dec 2017 **8 from VUT**, *first place*.

Sep 2017 **Red Diploma**, *for excellent results in Bachelor studies*.

Sep 2017 **Dean's award**, *for excellent results in Bachelor studies*.

Sep 2017 **Rector's award**, *for excellent results in Bachelor studies*.

Mar 2016 **EBEC Brno**, *second place*.

Interests

- Competitive programming
- Teaching and learning
- Personal development
- Dancing
- Travelling
- Foreign languages
- Philosophy
- Public speaking